

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November, 2013

CL 313: Hydrology and Groundwater Hydraulics

Date: 27th November, 2013

Day: Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is the importance of hydrological studies in water resources projects? 03
- (b) State the Darcy's Law for movement of ground water. Explain upconing of saline water. 04
- Q - 2 (a) Enlist the factors affecting groundwater flow. What is aquitard? 03
- (b) What are the needs of conjunctive use? 04
- (c) Define: Storage Coefficient, Porosity and Permeability. 03
- (d) It was observed in a field test that 2 hr 30 min was required for a tracer to travel from one well to another 20 m apart and the difference in their water surface elevations was 0.75 m. Samples of the aquifer between the wells indicated a porosity of 10%. Determine the permeability of the aquifer, seepage velocity and the Reynolds number for the flow, assuming an average grain size of 1 mm and ν at $27^\circ\text{C} = 0.009$ Stoke. 04

OR

- Q - 2 (a) Differentiate unconfined and confined aquifer. 03
- (b) Enlist methods of geophysical investigation of groundwater. Explain any one. 04
- (c) Define: transmissibility, aquiclude and aquifuge. 03
- (d) An aquifer of aerial extent of 125 km^2 is overlain by five strata as given below: 04

Strata	Thickness (m)	Kx (m/day)	Ky (m/day)
1 (Top)	2	2	0.31
2	3	1.5	0.27
3	1	2.3	0.045
4	4	0.22	0.09
5	2	0.036	0.021

If a 3.5 hr storm occurs producing a total rainfall 150 mm, estimate the recharge into the aquifer, assuming that the piezometric surface in the aquifer is at the bottom of layer 5th and that all layers are saturated.

- Q - 3 (a) The time-drawdown data from an observation well at 12.3 m from a pumped well is given in table below. The test well is pumped at the rate of 1150 lpm. Static water level in the test well (before pumping started) is 2.18 m. Determine the constants 'T' and 'S' by Chow's method. 09

Time – drawdown data:

Time 't' (min)	Depth of measuring point (m)	Time 't' (min)	Depth of measuring point (m)
0	2.18	18	2.69
1	2.42	22	2.71
2	2.42	28	2.72
3	2.46	35	2.75
4	2.50	45	2.82
6	2.55	55	2.83
8	2.59	65	2.86
10	2.63	80	2.87
14	2.67	100	2.92
		120	2.94

Drawdown Values:

Time 't' (min)	Drawdown 's' (m)	Time 't' (min)	Drawdown 's' (m)
1	0.24	22	0.53
2	0.24	28	0.54
3	0.28	35	0.57
4	0.32	45	0.64
6	0.37	55	0.65
8	0.41	65	0.68
10	0.45	80	0.69
14	0.49	100	0.74
18	0.51	120	0.76

- (b) Which prevention measures and methods have been taken to control the salt water intrusion? 02
- (c) Explain well losses. 03

SECTION – II

- Q - 4 (a) Differentiate between natural and artificial ground water recharge. What are the advantages of artificial ground water recharge? 03
- (b) What are the needs of ground water recharge? Describe the components of Roof Rain Water Harvesting System with neat sketch. 04
- Q - 5 (a) Explain: Unit Hydrograph (UH). What are the assumptions in UH theory? 03
- (b) A sub-basin has six numbers of rain gauges. Annual rainfalls recorded by the gauges are given below. Considering 10% error in the estimation of mean rainfall, calculate optimum number of gauges required for the sub-basin and check if present network is sufficient. 04

Rain gauge name	A	B	C	D	E	F
Annual rainfall (cm)	110.3	82.6	98.8	136.7	180.3	102.9

- (c) What are the different methods of measurement of evaporation? Also, explain, "Pan coefficient". 03
- (d) In a 105 minute storm, the rainfall observed in successive 15 min intervals were 3.0, 3.0, 9.0, 6.6, 1.2, 1.2 and 6.0 mm/hr. Assuming the ϕ - index value as 3 mm/hr and initial loss of 1 mm, determine the total rainfall and W-index for the storm. 04

OR

- Q - 5 (a) Compute the average depth of rainfall and volume of rain water using Thiessen polygon method for the data given below: 03

Station	A	B	C	D	E	F	G	H
Rainfall (mm)	40	48	38	55	49	37	25	40
Area of Polygon (km ²)	1.5	41.5	47.2	6.6	35.8	49.8	10.4	7.2

- (b) What are the different methods for flood frequency analysis for the estimation of flood discharge? Discuss any one of them. 04
- (c) What is S-hydrograph? How is it derived from a Unit hydrograph? 03
- (d) Find out the ordinates of a storm hydrograph resulting from a 3 hour storm with rainfall of 3, 4.5 and 1.5 cm during subsequent 3 hours intervals. The ordinates of 3-hr Unit Hydrograph (UH) for the catchment is given below: 04

Time (hr)	0	3	6	9	12	15	18	21	24	27	30	33	36
3-hr UH ordinates (cumecs)	0	90	200	350	450	350	260	190	130	80	45	20	0

Assume an initial loss of 5mm, infiltration index of 5 mm/hour and base flow of 20 cumecs.

- Q - 6 (a) Explain 'Recording rain gauge'. Enlist different types of recording rain gauges and discuss any one of them in detail. 04

- (b) For a river, the estimated flood peaks for two return periods by the use of Gumbel's method are as follows: 05

Return Period (Years)	Peak flood (m ³ /s)
100	435
50	395

What flood discharge in this river will have a return period of 1000 years?

- (c) Explain the following terms: 05
 (i) Direct runoff (ii) Base flow (iii) SPF (iv) Evapotranspiration (v) Hyetograph

Equations:

$$y_T = -[\ln.\ln.(T/(T-1))]$$

$$x_T = \bar{x} + K\sigma_{n-1}$$

$$K = (y_T - \bar{y}_n)/S_n$$